Exercise 11

Mittwoch, 28. Januar 2026 21:15

Exercise: Basic Controller for the Longitudinal Motion

Given the following state-space representation, design a basic controller for the Boeing 777.

$$\begin{pmatrix} \dot{V}^* \\ \dot{\gamma} \\ \dot{\alpha} \\ \dot{q} \end{pmatrix} = \begin{bmatrix} -0.01 & -0.49 & -0.046 & -0.001 \\ 0.11 & 0.0003 & 1.14 & 0.043 \\ -0.11 & -0.0003 & -1.14 & 0.957 \\ 0.1 & 0 & -15.34 & -3.00 \end{bmatrix} \begin{pmatrix} V^* \\ \gamma \\ \alpha \\ q \end{pmatrix} + \begin{bmatrix} 0.0002 & 0.014 \\ -0.059 & 0 \\ 0.059 & 0 \\ 6.56 & 0.13 \end{bmatrix} \cdot \begin{pmatrix} \eta \\ \delta_T \end{pmatrix}$$

The controller is supposed to augment the short-period eigenmode such that:

- the ideal damping $\zeta_{SP} = \frac{1}{\sqrt{2}}$ and
- the desired eigenfrequency $\omega_{0,SP} = 4{,}5\,\frac{rad}{s}$

are achieved.

a) Analyze the eigenmode characteristics of the aircraft without a controller.

1) Simplified State Space Model:

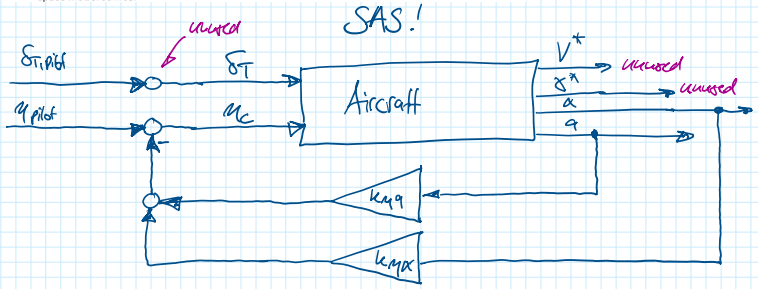
$$\begin{bmatrix} \dot{\alpha} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} Z_\alpha & Z_q+1 \\ M_\alpha & M_q \end{bmatrix} \begin{bmatrix} \alpha \\ q \end{bmatrix} + \begin{bmatrix} Z_{\delta_T} \\ M_{\delta_T} \end{bmatrix} \eta$$

$$A_{777} = \begin{bmatrix} -1.14 & 0.957 \\ -15.34 & -3.00 \end{bmatrix}, \quad B_{777} = \begin{bmatrix} 0.059 \\ 6.56 \end{bmatrix}$$

$\Rightarrow$ Eigenfrequency: $\omega_{0,SP}^2 = Z_\alpha M_q - M_\alpha (Z_q+1) = 4.25\,\frac{rad}{s}$; $\overset{!}{\Longleftrightarrow}$ Desired value $\omega_{0,SP,des} = 4.5\,\frac{rad}{s}$

$$\zeta = -\frac{\sigma}{\omega_0}\,\zeta_{SP} = -\frac{1}{2}\cdot\frac{Z_\alpha + M_q}{\omega_{0,SP}} = 0.49 \quad \overset{!}{\Longleftrightarrow} \quad \zeta_{SP,des} = \frac{1}{\sqrt{2}}$$

b) Propose a suitable controller structure, for which the measurements of the states in the above state space model suffice.



$$\eta_c = \eta_{pilot} - k_{\alpha\alpha}\cdot\alpha - k_{\alpha q}\cdot q$$

$$\begin{bmatrix} \dot{\alpha} \\ \dot{q} \end{bmatrix} = \begin{bmatrix} Z_\alpha & Z_q+1 \\ M_\alpha & M_q \end{bmatrix} \begin{bmatrix} \alpha \\ q \end{bmatrix} + \begin{bmatrix} Z_{\delta_T} \\ M_{\delta_T} \end{bmatrix} \cdot \eta_c = \begin{bmatrix} Z_\alpha & Z_q+1 \\ M_\alpha & M_q \end{bmatrix} \begin{bmatrix} \alpha \\ q \end{bmatrix} + \begin{bmatrix} Z_{\delta_T} \\ M_{\delta_T} \end{bmatrix} (\eta_{pilot} - k_{\alpha\alpha}\cdot\alpha - k_{\alpha q}\cdot q)$$

Feedback Matrix

$$= \begin{bmatrix} Z_\alpha & Z_q+1 \\ M_\alpha & M_q \end{bmatrix} \begin{bmatrix} \alpha \\ q \end{bmatrix} + \begin{bmatrix} Z_{\delta_T}(\eta_{pilot} - k_{\alpha\alpha}\cdot\alpha - k_{\alpha q}\cdot q) \\ M_{\delta_T}(\eta_{pilot} - k_{\alpha\alpha}\cdot\alpha - k_{\alpha q}\cdot q) \end{bmatrix} = \begin{bmatrix} Z_\alpha & Z_q+1 \\ M_\alpha & M_q \end{bmatrix} \begin{bmatrix} \alpha \\ q \end{bmatrix} - \begin{bmatrix} Z_{\delta_T}k_{\alpha\alpha} & Z_{\delta_T}k_{\alpha q} \\ M_{\delta_T}k_{\alpha\alpha} & M_{\delta_T}k_{\alpha q} \end{bmatrix} \begin{bmatrix} \alpha \\ q \end{bmatrix} + \begin{bmatrix} Z_{\delta_T} \\ M_{\delta_T} \end{bmatrix} \eta_{pilot}$$

$$= \begin{bmatrix} Z_\alpha - Z_{\delta_T}k_{\alpha\alpha} & Z_q+1 - Z_{\delta_T}k_{\alpha q} \\ M_\alpha - M_{\delta_T}k_{\alpha\alpha} & M_q - M_{\delta_T}k_{\alpha q} \end{bmatrix} \begin{bmatrix} \alpha \\ q \end{bmatrix} + \begin{bmatrix} Z_{\delta_T} \\ M_{\delta_T} \end{bmatrix} \cdot \eta_{pilot}$$

$\rightarrow A_{CL}$ Matrix

c) Determine the required controller gains such that the above-mentioned requirements for the short-period are fulfilled.

$$A_{CL}: \begin{bmatrix} Z_\alpha^* & Z_q^*+1 \\ M_\alpha^* & M_q^* \end{bmatrix} \quad \text{with}$$

$$Z_\alpha^* = Z_\alpha - Z_{\delta_T}k_{\alpha\alpha}$$
$$M_\alpha^* = M_\alpha - M_{\delta_T}k_{\alpha\alpha}$$
$$Z_q^* = Z_q - Z_{\delta_T}k_{\alpha q}$$
$$M_q^* = M_q - M_{\delta_T}k_{\alpha q}$$

$\rightarrow$ Compute $\det(sI - A_{CL})$ $\rightarrow$ obtain characteristic polynom & solve for coefficients

$$\omega_{0,SP,des} = \sqrt{Z_\alpha^* M_q^* - M_\alpha^*(Z_q^*+1)} = \sqrt{(Z_\alpha - Z_{\delta_T}k_{\alpha\alpha})(M_q - M_{\delta_T}k_{\alpha q}) - (M_\alpha - M_{\delta_T}k_{\alpha\alpha})(Z_q - Z_{\delta_T}k_{\alpha q} + 1)} = 4.5\,\frac{rad}{s}$$

$$\zeta_{SP,des} = -\frac{1}{2}\frac{Z_\alpha^* + M_q^*}{\omega_{0,SP,des}} = -\frac{1}{2}\cdot\frac{(Z_\alpha - Z_{\delta_T}k_{\alpha\alpha}) + (M_q - M_{\delta_T}k_{\alpha q})}{\omega_{0,SP,des}} = \frac{1}{\sqrt{2}}$$

$$\zeta_{SP,des} = -\frac{1}{2}\frac{z_\alpha^* + M_q^*}{\omega_{0,SP,des}} = -\frac{1}{2}\cdot\frac{(z_\alpha - z_{\eta_q}u_{\eta_\alpha}) + (M_q - M_{\eta_q}u_{\eta_\alpha})}{\omega_{0,SP,des}} = \frac{1}{\sqrt{2}}$$

→ Solve for gains $u_{\eta_q}, u_{\eta_\alpha}$

$$\zeta_{SP,des} = -\frac{1}{2}\cdot\frac{z_\alpha + M_q - M_\eta u_{\eta_q}}{\omega_{0,SP,des}} \rightarrow u_{\eta_q} = \frac{2\cdot\zeta_{SP}\cdot\omega_{0SP,des} + z_\alpha + M_q}{M_\eta} = \boxed{0.3389;}$$

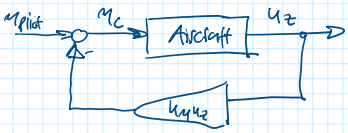
($z_\eta = 0$)

$$\omega_{0,SP,des}^2 = z_\alpha(M_q - M_q\cdot u_{\eta_q}) - (M_\alpha - M_\eta u_{\eta\alpha})(z_q + 1) = \ldots \Rightarrow u_{\eta\alpha} = \frac{\omega_{0SP,des}^2 - z_\alpha(M_q - M_q\cdot u_{\eta_q}) + M_\alpha(z_q+1)}{M_\eta(z_q+1)} = \boxed{-0.0644;}$$

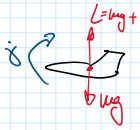
How big is the error compared to the unsimplified equation? Use $u_{\eta\alpha}, u_{\eta_q}$ corrected → $\omega_{0SP,des,full} = 4.46\ \frac{rad}{s} \sim 0.84\ \frac{rad}{s} \sim 0.9\%$

$\zeta_{SP,full} = 0.707 \sim 0.0001 \sim 0.1\%$

d) Propose an alternative control structure that accounts for inaccurate measurements of the angle of attack. Compute its required controller gains.



- increase/decrease the natural frequency of the SP
- relation damping remains constant for inaccurate feedback gain



$$\dot\gamma = \frac{1}{mV}(L+mg) = \frac{g}{V}\left(\frac{L-mg}{mg}\right) = \frac{g}{V}(u_z - u_{z0}) = \frac{g}{V}\delta u_z$$

$$\rightarrow \delta u_z = \frac{V}{g}\cdot\dot\gamma = \frac{V}{g}(\dot\theta - \dot\alpha) = \frac{V}{g}(q - \dot\alpha)$$

This derivation applies only to small angles and deviations from level flight

$\delta\dot\alpha = z_\alpha\cdot\delta\alpha + \delta q + z_{\eta_q}\delta\eta_c$

$$\delta u_z = \frac{V_0}{g}\left(\delta q - (z_\alpha\delta\alpha + \delta q + z_{\eta_q}\delta\eta_c)\right) = -\frac{V_0}{g}(z_\alpha\delta\alpha + z_{\eta_q}\delta\eta_c) = \delta u_z$$

$$\frac{-\frac{g}{V_0}\delta u_z + z_{\eta_q}\eta_c}{z_\alpha} = \delta\alpha$$

$$\Rightarrow \delta\dot u_z = -\frac{V_0}{g}(z_\alpha\cdot\delta\dot\alpha + z_{\eta_q}\cdot\delta\dot\eta_c)$$

$$\delta\dot u_z = -\frac{V_0}{g}\left(z_\alpha(z_\alpha\delta\alpha + \delta q + z_{\eta_q}\delta\eta_c) + z_{\eta_q}\delta\dot\eta_c\right)$$

$\delta u_z = -\frac{V_0}{g}(z_\alpha\delta\alpha + z_{\eta_q}\delta\eta_c)$

$$\boxed{\delta\dot u_z = z_\alpha\delta u_z - \frac{V_0}{g}z_\alpha\delta q - \frac{V_0}{g}z_{\eta_q}\delta\dot\eta_c}$$

— with $\delta u_z, \delta q$ and $\delta\dot\eta_c$ will replace $\delta\dot\alpha$-Equation

The new State Space Model: $z_\eta \approx 0$

$$\begin{bmatrix}\delta\dot u_z\\ \dot q\end{bmatrix} = \begin{bmatrix} z_\alpha & -\frac{V_0}{g}z_\alpha \\ M_\alpha\frac{g}{V_0 z_\alpha} & \end{bmatrix}\begin{bmatrix}\delta u_z\\ q\end{bmatrix} + \begin{bmatrix}0\\ M_\eta\end{bmatrix}\cdot\eta_c$$

$$\delta\alpha = \frac{\frac{g}{V_0}\cdot\delta u_z - z_{\eta_q}\delta\eta_c}{z_\alpha} = \frac{g}{V_0\cdot z_\alpha}\cdot\delta u_z \quad (z_\eta \approx 0)$$

$$M_\alpha\cdot\alpha = \left(M_\alpha\cdot\frac{g}{V_0\cdot z_\alpha}\right)\delta u_z$$

The new control law: $\eta_c = \eta_{pilot} - u_{\eta_q}\cdot q - u_{\eta_{u_z}}\cdot\delta u_z$

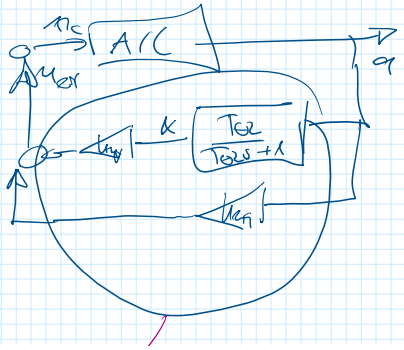
2) Upside Feedback Structure: Lead-Lag-Filter to estimate α from q

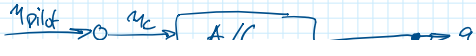
$$\dot\alpha = z_\alpha\cdot\alpha + q$$

$$s\alpha = z_\alpha\cdot\alpha + q \rightarrow \text{TF: } G_{\alpha q}(s) = \frac{\alpha(s)}{q(s)} = \frac{1}{s - z_\alpha} = \frac{-1/z_\alpha}{-1/z_\alpha\cdot s + 1} = \frac{T_{\theta 2}}{T_{\theta 2}s + 1}$$

$$\Rightarrow \delta\alpha = \frac{T_{\theta 2}}{T_{\theta 2}\cdot s + 1}\cdot\delta q \quad \text{with } T_{\theta 2} = -\frac{1}{z_\alpha}$$

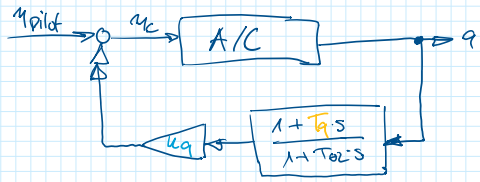
Model dependent → Major source of uncertainty





$T_{62} \cdot s + 1$

uncertainty



1) Integrate lead lag filter into control law:

$$n_c = n_{pilot} - (u_{nq} \cdot \delta_q + u_{nx} \cdot \delta_x)$$

$$= n_{pilot} - \left(u_{nq} \cdot \delta_q + u_{nx} \cdot \frac{T_{62}}{T_{62} \cdot s + 1} \cdot \delta_q\right) = \ldots =$$

$$= n_{pilot} - \left(\underbrace{u_{nq} + u_{nx} \cdot T_{62}}_{u_q} \cdot \frac{\frac{u_{nq} T_{62}}{u_{nq} + u_{nx} \cdot T_{62}} \cdot s + 1}{1 + T_{62} \cdot s} \right) \delta_q$$

T_q

Combine both feedback paths into one!

$$\Rightarrow T_{62} = -\frac{1}{z_a} = \frac{1}{1.14}$$, insert computed values from previous attach u_{nq}, u_{nx}

$$T_q = 1.043$$

$$u_q = 0.2851$$